# Gregor Ethan Rubin Posadas, M.S., E.I., ENV SP

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## EDUCATION

**Doctor of Philosophy in Civil and Environmental Engineering** — In Progress (Expected 2031)
University of California, Berkeley — Berkeley, California
Environmental Engineering Program | Designated Emphasis in Development Engineering
Advisor: Dr. Kara L. Nelson | Co-advisor: Dr. Maya Carrasquillo

**Master of Science in Civil Engineering** — December 2025
Boise State University — Boise, Idaho
*Thesis: Evaluation of High-Concentration Dissolved Oxygen Delivery on Wastewater Treatment Plant Performance*
GPA: 4.0 | Tau Beta Pi Engineering Honor Society; Chi Epsilon Civil Engineering Honor Society

**Bachelor of Science in Civil Engineering** — December 2023
Boise State University — Boise, Idaho
Cum Laude | GPA: 3.74 | Dean's List (4 semesters)

## RESEARCH EXPERIENCE

**Research Fellow, Water Infrastructure and Policy** — September 2025 – August 2026
Water Finance Exchange (WFX), Inc. — Washington, D.C. / Remote

- Conducted research on the adoption and implementation of new technologies for water infrastructure systems in small and rural communities across the United States.
- Performed purposive semi-structured stakeholder interviews and matrix coding analysis to surface barriers to effective water infrastructure-related policy implementation.
- Analyzed Aquatic Resource Alteration Permits (ARAPs) for water systems in Tennessee's Duck River watershed to identify opportunities for utility regionalization and consolidation.
- Developed a regionalization cost-estimate analysis by reviewing historical State Revolving Fund (SRF) funding requests from water systems across Tennessee, benchmarking the cost of infrastructure such as water treatment plant expansions, pumps, water lines, and water meters.

**Research Manager** — February 2026 – August 2026
Boise State University, Department of Civil Engineering — Remote

- Supervised research projects for Dr. Sondra Miller's research group, coordinating priorities and timelines across multiple concurrent studies.
- Mentored graduate and undergraduate students in research writing and the development of conference presentations.
- Developed grant proposals, data analysis reports, and peer-reviewed research publications.

**Graduate Research Assistant** — December 2023 – December 2025
Boise State University, Department of Civil Engineering — Boise, Idaho

- Performed comprehensive wastewater sampling and analysis (BOD, COD, nitrogen compounds, phosphorus, TSS, ATP) on a fully operational treatment facility.
- Conducted feasibility testing on novel retrofittable wastewater technology designed to improve bioreactor oxygenation and contaminant removal.
- Developed statistical and machine-learning approaches to evaluate relationships among dissolved oxygen, biomass activity, and sludge settleability.
- Mentored project-track graduate students through the journal publication process.

**Undergraduate Research Assistant** August 2023 – December 2023
Boise State University, Department of Civil Engineering Boise, Idaho

- Developed Python programs to support environmental laboratory experiments, including data analysis for settling column tests.
- Managed day-to-day operations of the Boise State University Environmental Engineering lab, maintaining equipment, safety protocols, and inventory.

## TEACHING EXPERIENCE

**Lab Instructor — CE 323: Environmental and Fluids Laboratory** August 2025 – December 2025
Boise State University, Department of Civil Engineering Boise, Idaho

- Led hands-on lab sessions on water and wastewater quality testing (BOD, turbidity, pH, dissolved oxygen) and fluid mechanics.
- Evaluated lab reports and student performance to ensure mastery of experimental methods and engineering analysis.
- Taught students to analyze and interpret experimental data using statistical and graphical methods.

**Teaching Assistant — CE 220: Coupled Environmental Processes** January 2025 – April 2025
Boise State University, Department of Civil Engineering Boise, Idaho

- Graded assignments and exams, providing constructive feedback to improve student comprehension and analytical skills.
- Hosted weekly office hours offering one-on-one and small-group support on water and wastewater treatment concepts.

**Guest Lecturer — CE 320: Environmental Engineering** October 2024
Boise State University, Department of Civil Engineering Boise, Idaho

- Introduced methods for evaluating particle settling velocity distributions and constructing isoconcentration contour plots using beginner-friendly Python via Google Colab.

## PUBLICATIONS

*Peer-reviewed journal articles and conference proceedings (Posadas, G. bolded):*

**Posadas, G.**, & Bonacquist-Currin, M. (2026). *Barriers to technology adoption in small and rural water systems: A multi-stakeholder interview study*. Manuscript in preparation for submission to Engineering Research Express.

Henson, R. L., **Posadas, G.**, & Miller, S. M. (2026). *Toward ATP-informed aeration control in wastewater treatment: A perspective*. Environmental Research Communications. https://doi.org/10.1088/2515-7620/aeb0f8

**Posadas, G.**, & Miller, S. (2026). *Wastewater treatment in Idaho in the face of emerging contaminants: Policy lag, utility burdens, and infrastructure readiness.* Environmental Research Communications. https://doi.org/10.1088/2515-7620/ae420a

**Posadas, G.**, Davis, K., & Miller, S. (2025). *High-concentration dissolved oxygen delivery in wastewater treatment: Barriers between innovation and adoption.* Environmental Research: Water. https://doi.org/10.1088/3033-4942/adffab

**Posadas, G.**, Davis, K., & Miller, S. (2025). *Liquid and solid phase effects of elevated bioreactor dissolved oxygen.* Proceedings of the 13th World Congress on Water Resources and Environment (EWRA 2025).

**Posadas, G.**, Davis, K., & Miller, S. (2025). *Assessing the impacts of a novel dissolved oxygen infusion system on wastewater treatment plant performance.* Environmental Research Communications. https://doi.org/10.1088/2515-7620/addbaf

**Posadas, G.**, Baker, J. C., Ryan, A. S., & Miller, S. M. (2025). *Resiliency through applied research into emerging constituents*. Water Resources Management. https://doi.org/10.1007/s11269-025-04140-7

**Posadas, G.**, & Miller, S. (2025). *Dissolved oxygen influences on sludge settleability: Mediating role of the volatile-to-total suspended solids ratio.* Accepted for publication in the 10th Asia Conference on Environment and Sustainable Development (ACESD 2025) conference proceedings.

## CONFERENCE PRESENTATIONS

**BahaWatch: Low-cost, open-source, locally-operated flood sensing for Philippine communities (and beyond)** October 2026
Anglo, N., **Posadas, G.**, & Groeschel, T. — IEEE Global Humanitarian Technology Conference (GHTC 2026), Boulder, CO — Poster Presentation (1st Place, Student Poster Competition)

**ATP-based monitoring to indicate microbial performance in wastewater treatment plants** July 2026
Smith, L., **Posadas, G.**, Henson, R., & Miller, S. — 2026 Idaho Conference on Undergraduate Research (ICUR), Boise State University, Boise, ID — Poster Presentation

**Understanding wastes from Idaho beverage production** April 2026
Parten, J., **Posadas, G.**, & Miller, S. — 2026 Undergraduate Research Showcase (URS), Boise State University, Boise, ID — Poster Presentation

**Real-time opto-electrochemical PFAS and pharmaceutical sensing in water** March 2026
Miller, S. M., Crockett, L. M., **Posadas, G. R.**, Omotowa, B., & Campbell, K. A. — DoW Applied Innovation Workshop, Washington Hilton, Washington, D.C. — Poster Presentation

**Dissolved oxygen influences on sludge settleability: Mediating role of the volatile-to-total suspended solids ratio** November 2025
10th Asia Conference on Environment and Sustainable Development (ACESD 2025), Fukuoka, Japan — Oral Presentation

**Assessing the impacts of a novel dissolved oxygen infusion system on wastewater treatment plant performance** September 2025
Pacific Northwest Clean Water Association Annual Conference (PNCWA 2025), Portland, OR — Oral Presentation

**Liquid and solid phase effects of elevated bioreactor dissolved oxygen** June 2025
13th World Congress on Water Resources and Environment (EWRA 2025), Palermo, Italy — Oral Presentation

**Evaluation of high-concentration dissolved oxygen delivery on wastewater treatment plant performance** November 2024
S-STEM Scholars & PI Meeting (AAAS & NSF), Chicago, IL — Poster Presentation

## HONORS & AWARDS

**1st Place, Student Poster Competition** October 2026
16th IEEE Global Humanitarian Technology Conference (GHTC 2026), Boulder, CO — for "BahaWatch," with N. Anglo and T. Groeschel

**Chancellor's Fellowship** August 2026
University of California, Berkeley

**IOP Publishing Outstanding Reviewer Award** March 2026
Institute of Physics (IOP) Publishing

**Semi-finalist, Fulbright U.S. Student Program (Singapore)** 2026
U.S. Department of State, Bureau of Educational and Cultural Affairs

**Inaugural MLK Jr. Living Legacy Committee Graduation Stole Recipient** May 2025
Boise State University MLK Jr. Living Legacy Committee

**Spring 2025 Graduating Student Leader Award (Graduate)** May 2025
Boise State University Student Involvement and Leadership Center

**Catalyzing Advocacy in Science and Engineering (CASE) Workshop AAAS Sponsorship Award** March 2025
American Association for the Advancement of Science (AAAS)

**United States Army Achievement Medal (AAM)** September 2024
U.S. Army Reserve, 321st Engineer Battalion

**Martin Luther King Jr. and Meredyth Burns Scholar** April 2024
Boise State University

**Commitment to Community Award Nominee** April 2024
Boise State University Division of Student Affairs

**ASCE New Face of Civil Engineering — Collegiate Honoree** February 2024
American Society of Civil Engineers (ASCE)

**NSF S-STEM Stellar Engineering Students Graduate Program Scholar (SEnS GPS)** December 2023
Boise State University Department of Civil Engineering

**Graduating Student Leader Award (Undergraduate)** December 2023
Boise State University Student Involvement and Leadership Center

**Fall 2023 Dean's List with High Honors** December 2023
Boise State University College of Engineering

**Idaho Opportunity Scholar** July 2023
Idaho State Board of Education

**391st Engineer Company Commander's Coin** April 2023
U.S. Army Reserve, 391st Engineer Company

**Spring 2023 Dean's List with High Honors** May 2023
Boise State University College of Engineering

**Spring 2021 Dean's List with Honors** May 2021
Boise State University College of Engineering

**Donald E. and Edithe L. Barclay Civil Engineering Scholar** April 2021
Boise State University Department of Civil Engineering

**Fall 2019 Dean's List with Highest Honors** December 2019
Boise State University College of Engineering

## CERTIFICATIONS & LICENSES

**Engineer Intern (E.I.) — Credential E-22940** January 2024
Idaho State Board of Professional Engineers and Land Surveyors

**Envision Sustainability Professional (ENV SP) — Credential 59936** January 2024
Institute for Sustainable Infrastructure

**IOP Trusted Reviewer Status** October 2025
Institute of Physics (IOP) Publishing

**Fundamentals of Engineering (FE) Exam — Passed** December 2023
NCEES

## ADDITIONAL PROFESSIONAL EXPERIENCE

**Wastewater Data Analyst Consultant** August 2025 – July 2026
BioSolutions, U.S.A. Greenleaf, Idaho

- Conduct comparative measurements of activated-sludge dissolved oxygen concentrations to analyze oxygenation/aeration technology.
- Monitor treatment-unit process technologies' power consumption for efficiency analyses.

**Community Climate Action Committee Member** February 2025 – December 2025
City of Boise Boise, Idaho

- Provided input on city policies integrating sustainability, resilience, and climate adaptation into urban planning and infrastructure.
- Contributed to the city's Community Climate Action Plan and advocated for youth involvement in climate initiatives.

**Combat Engineer** August 2019 – July 2025
U.S. Army Reserve, 391st Engineer Company Boise, Idaho

- Performed engineering calculations for obstacle emplacement and explosive demolitions, ensuring precision and safety in military operations.
- Operated and maintained military vehicles (M113A1, HMMWV, AVLM/AVLB, HEMTT, LMTV) and trained soldiers on radio (SINCGARS) and navigation (DAGR) systems.

**Marketing and Graphic Design Delegate** November 2019 – January 2025
Boise State University MLK Jr. Living Legacy Committee Boise, Idaho

- Organized the annual MLK Day march in Boise and other community engagement events raising awareness of civil rights and social justice.
- Established organizational brand standards and designed a cohesive suite of digital and print marketing materials.

**Air Quality Monitoring Short-Term Consultant** March 2024 – April 2024
ADEC Environmental Meridian, Idaho

- Assessed a pilot air-quality monitoring technology by comparing measurement capability against established EPA methods.
- Conducted daily ambient air-quality monitoring and prepared particulate-matter (PM) samples for laboratory analysis.

**Civil Engineering Intern** May 2023 – September 2023
Stantec Boise, Idaho

- Developed operations and maintenance (O&M) manuals for municipal wastewater treatment plants, ensuring compliance with operational standards.
- Finalized construction drawings using AutoCAD, Autodesk Civil 3D, and Bluebeam Revu, and prepared Statements of Qualifications (SOQs).
- Designed databases to track project pursuits and prepared site-condition exhibits and background-study reviews.

**Lead Graphic Designer** August 2020 – May 2023
Dice & D-Pads Quezon City, Philippines

- Assigned responsibilities and weekly agendas for the graphic design team, ensuring clear delegation and efficient workflows.
- Produced marketing materials, videos, and social media content using Adobe Creative Suite, Canva, and DaVinci Resolve.

**Facilities, Operations, and Maintenance (FOM) Student Assistant** May 2021 – January 2022
Boise State University Boise, Idaho

- Reviewed and updated university building plans for the department's online building database.
- Liaised between building managers and maintenance personnel to resolve facilities issues.

**Help Desk Zone Support Analyst** October 2019 – December 2019
Boise State University, Office of Information Technology (OIT) Boise, Idaho

- Provided in-person IT support to faculty and students, troubleshooting classroom technology and campus-wide services.
- Maintained lectern systems and managed equipment loans while documenting activity in the IT ticketing system.

## ACADEMIC & DESIGN PROJECTS

**Sustainable Energy Simulation and Analysis for Boise State University** January 2024 – May 2024
Evaluated campus solar energy potential using PVSketch; produced economic projections and $CO_2$ offset analysis in coordination with the Boise State Office of Sustainability.

**Project Proposal Development — Clearwater Engineering** August 2023 – December 2023
Led a team responding to an RFP for a fish-barrier removal program; secured engineering partners and ranked first of four competing teams.

**Municipal Wastewater Treatment Plant Design** January 2023 – May 2023
Designed a theoretical municipal WWTP discharging to the Big Wood River — pump station, grit chambers, clarifiers, and UV disinfection — under IDAPA, HI, and GLUMRB standards; authored the Preliminary Engineering Report.

**ASCE Timber Strong Competition** January 2023 – April 2023
Helped design and construct a two-story timber structure for the ASCE Student Symposium (Boise State Civil Engineering Club).

**Subdivision Construction Drafting** January 2021 – May 2021
Created construction drawings for a Western Idaho subdivision, including roadway design, sewer/water line placement, and grade planning (Autodesk Civil 3D).

## SOFTWARE DEVELOPMENT PROJECTS

**Wastewater Treatment Plant Statistical Framework and Visualization Tool** February 2025
github.com/gregor-posadas/wwtp-network-visualization

- Built a Python-based tool for wastewater data analysis using multiple-correlation methods and interactive visualizations.

**Settling Column Isoremoval Curve Generator** December 2024
isoremoval.streamlit.app

- Developed an open-access Python learning tool that generates isoremoval curves from raw settling column data; adopted in an Environmental Engineering course.

## PUBLIC SPEAKING & OUTREACH

**Featured Pinoy Scientist** May 2025
pinoyscientists.com/scientist/gregor-ethan-posadas/

- Wrote and published daily social media posts on Philippine infrastructure and the role of Filipino scientists and engineers in national development.

**Student Keynote Speaker — Fall 2023 Asian and Pacific Islander (API) Stole Ceremony** December 2023
Hosted by the Boise State Student Equity Center (SEC) Boise, Idaho

- Delivered a keynote reflecting on cultural belonging and Filipino heritage.

**Storyteller — The Story Collider: Veterans in STEM** November 2023
storycollider.org/stories/2024/3/6/full-circle-stories-about-going-back-to-the-start

- Shared a personal narrative on the intersection of military service and a career in engineering.

## SERVICE & VOLUNTEERING

**Student Volunteer** April 2024 – Present
ASME EFx Event — Boise State University (Tau Beta Pi)

**Student Volunteer — Tree Planting** April 2024
Arbor Day Foundation

**Volunteer** November 2023
Rake Up Boise (Boise State Tau Beta Pi, Idaho Gamma Chapter)

## PROFESSIONAL AFFILIATIONS & LEADERSHIP

**American Association for the Advancement of Science (AAAS)** April 2024 – Present
Student Member

**American Society of Civil Engineers (ASCE)** October 2022 – Present
Student Member

**Chi Epsilon Civil Engineering Honor Society** May 2023 – May 2025
Vice President (2024–25); Recording Marshal (2023–24)

**Water Environment Federation (WEF)** August 2024 – May 2025
Student Member

**Tau Beta Pi Engineering Honor Society** April 2023 – May 2024
Chapter Secretary

**ASBSU Sustainability Committee** February 2024 – May 2024
Student Member At-Large

**Associated Students of Boise State University (ASBSU)** February 2024 – May 2024
Review Board Chief

## TECHNICAL SKILLS

**Engineering & Design Software:** AutoCAD, Autodesk Civil 3D, Navisworks, Autodesk BIM 360, ProjectWise, Bluebeam Revu, SketchUp, PVSketch

**Programming & Data:** Python, MATLAB, statistical data analysis, machine learning, data visualization

**Laboratory & Analytical:** Wastewater sampling and analysis (BOD, COD, TSS, nitrogen, phosphorus, ATP); DO/pH/temperature multi-probe field methods

**Productivity & Creative:** Microsoft Office (Excel, Word, PowerPoint), Adobe Creative Suite, Canva, DaVinci Resolve

**Professional:** Research and technical writing, proposal writing, project management, grant development, stakeholder engagement

## LANGUAGES

English (native/bilingual) • Tagalog (native/bilingual)

## REFERENCES

Available upon request.